

Design and Evaluation of a Deterministic Query Orchestrator for a Distributed Hybrid Retrieval System

Msc Computer Science



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Software Project Report
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Abstract

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1. Introduction

This report begins by establishing the way this project aims to contribute to addressing the challenges involved in the enterprise retrieval systems. It reviews the related work and then moves on to explain the methodology adopted in design of the orchestrator, followed by the results recorded in the use of the orchestrator against a gold standard benchmark, and lastly, the evaluation and discussion of those results.

1.1 Problem Statement

The advent of large language models (LLMs) [1] has opened up numerous applications where a conversation can be had with an LLM using the knowledge encoded in the parameters of the model during pre-training. While fascinating, this has a limitation when it comes to their use within enterprises. Enterprise data is often gated and hence is not included in the pre-training of the LLMs. This can lead to hallucinations where the LLM confidently responds to queries with false information where it does not have innate knowledge. Retrieval augmented generation (RAG) [2], while based on techniques long known in the domains of search and recommendation systems, has been found to be a reliable way to ground the responses of LLMs in relevant and recent information to reduce hallucinations [3].

RAG is extensively used to provide gated enterprise knowledge to LLMs to ground their responses. Depending on the shape of information, the enterprise knowledge is often distributed across several knowledge stores such as vector stores, document stores, relational tables and graph databases. As this knowledge grows, the knowledge retrieval systems start taking the form of a distributed system where these knowledge stores needs to be either partitioned or replicated or both depending on whether consistency or availability is targeted [4].

Also, in enterprise environments, deterministic orchestration is often preferred because retrieval decisions are predictable and auditable, allowing system designers to understand why particular data sources were queried for a given request. However, the performance implications of adopting deterministic orchestration strategies in distributed hybrid retrieval systems remain insufficiently understood.

Little work evaluates how different deterministic orchestration strategies influence end-to-end retrieval latency and retrieval quality when semantic and structured retrieval must be coordinated across distributed knowledge stores. Consequently, practitioners have limited evidence regarding which orchestration strategies provide the best trade-off between latency and retrieval quality under different sharding

configurations.

Lastly, in any information retrieval system, it is important to maintain a balance between the speed and accuracy of the response served to the user to maintain a higher degree of user delight [5]. This project addresses these concerns through its specific contributions.

1.2 Contribution

Review of related work demonstrates significant progress in semantic retrieval, distributed RAG, search indexes for relational and vector stores, distributed systems, query planning, and hybrid retrieval [6]. Existing work has addressed both individual retrieval components and their coordination, including resource discovery, selection, and reasoning over text and tables.

However, the reviewed work in this report does not establish how alternative deterministic orchestration strategies compare within a common sharded system supporting structured relational, dense semantic and sparse lexical retrieval. Here, deterministic strategies are those that follow explicitly defined routing and coordination rules for a given input and system state.

This project explores this alternative by designing and experimentally evaluating alternative routing and orchestration strategies under a sharded knowledge store using established information retrieval and distributed systems performance metrics.

Also, in the information-retrieval literature, hybrid retrieval commonly refers to combining sparse lexical and dense semantic retrieval signals into a unified ranking. The system developed in this project does not perform this form of score or rank fusion. Instead, it is a query orchestrator for a hybrid retrieval system (HRS) supporting structured relational, dense semantic, and sparse lexical retrieval over aligned distributed data using a unified orchestration. Here, aligned data refers to corresponding information represented across the different retrieval stores.

1.3 Aim and Objectives

The aim of this project is to design and implement a deterministic query orchestrator for a distributed hybrid retrieval system, and experimentally evaluate how routing and execution strategies affect retrieval latency and effectiveness across defined workloads and sharding configurations

To achieve this aim, the project has the following objectives:

1. Design and implement a query orchestration framework, exposed through a

stateless hypertext transfer protocol (HTTP) application programming interface (API) supporting structured relational, dense semantic and sparse lexical retrieval over aligned distributed data.

2. Implement and compare deterministic shard-routing strategies, including broadcast and selective shard querying, and execution strategies, including filter-then-search and search-then-filter, for applicable query types.
3. Establish a reproducible evaluation benchmark comprising development and held-out test queries, with documented ground-truth results and relevance judgements appropriate to each query type.
4. Evaluate the strategies under controlled workloads and sharding configurations using mean, P95 and P99 retrieval latency [7], precision, recall and F1-score [5], and NDCG@k for ranked results [8]. For known-item queries, measure success as the proportion of queries for which the designated target appears within the top k results.
5. Analyse the trade-offs between latency and retrieval effectiveness, identifying the workload and configuration conditions under which each strategy offers advantages, and explaining the limitations of the findings. quality.

The following sections justify the methods selected to address these objectives, describe their implementation, and evaluate the extent to which the objectives were achieved.

2. Literature Review

This project lies at the intersection of RAG, natural language to SQL (NL2SQL) [9], federated search, vector search, relational databases and distributed systems. Consequently, the literature review draws upon contributions from each of these domains to motivate the proposed architecture and identify the research gap.

2.1 RAG

RAG has received considerable attention from the research community. Xu et al. [10] propose a distributed RAG (DRAG) system using a peer-to-peer (P2P) network of knowledge stores and local LLMs. Their healthcare-focused approach introduces a novel algorithm that retrieves information locally before querying neighbouring peers. While the system is distributed, it relies solely on vector embeddings as grounding information for LLMs and therefore does not address an HRS, where structured data must also be queried. Similarly, Guerraoui et al. [11] acknowledge the distributed nature of modern retrieval systems and propose RAGROUTE, a neural network classifier that predicts which knowledge store should be queried. Although this reduces unnecessary communication, the routing model is probabilistic and requires retraining whenever the underlying knowledge stores are updated. Recognising that enterprise retrieval extends beyond embeddings, Yu et al. [12] propose TableRAG, which embeds relational table schemas alongside textual content so that retrieved schema embeddings can guide subsequent SQL queries. While this demonstrates the value of heterogeneous retrieval, the paper does not discuss schema evolution and how very large distributed tables are handled in practice.

2.2 NL2SQL

As structured data becomes an increasingly important component of HRS, the quality of NL2SQL translation has a direct impact on retrieval quality. Existing approaches can broadly be categorised as rule-based, neural and hybrid methods. Shalaan et al. [13] propose Guided SQL (GSQL), which combines rule-based processing, natural language processing and vector embeddings to improve SQL generation. SQLizer adopts a different approach by first constructing a query sketch before iteratively materialising it against the database schema using CoreNLP and Word2Vec [14]. More recently, Liu et al. [9] observe that although rule-based systems provide explainability, they struggle with adaptability and complex schemas, while neural approaches remain constrained by training data. Consequently, the field has shifted towards LLM-based methods and fine-tuning techniques that demonstrate stronger generalisation across complex database schemas [9]. Given this direction, implementing a custom NL2SQL engine would contribute relatively little to the

primary aim of this project, which is the design and evaluation of query orchestration rather than query translation itself.

2.3 Resource Selection

Once queries have been translated, the retrieval system must determine which resources should be searched. Federated search addresses this problem by providing a unified interface through which multiple independent collections can be searched and their results combined [15]. Garba et al. [15] identify CORI [16] and ReDDE [17] as two of the most influential resource selection techniques. Callan et al. [16] use probabilistic inference networks to estimate the likelihood that a collection contains relevant information. Here, CORI relies primarily on term frequency (TF) and inverse document frequency (IDF), limiting its ability to capture semantic similarity. Si et al. [17] improve resource selection by estimating collection sizes and emphasise that reducing communication between distributed collections becomes increasingly important as the number of knowledge stores grows. Although these approaches were developed for keyword-based search rather than modern heterogeneous retrieval systems, these studies establish that resource selection remains a fundamental challenge in distributed retrieval.

2.4 Vector Search

Within a selected resource, semantic retrieval identifies relevant items by representing documents and queries as vector embeddings, allowing similarity to be measured using functions such as cosine similarity or Euclidean distance rather than exact keyword matching [4]. However, efficiently searching large vector collections presents another challenge. Flat indexes provide exact retrieval but exhibit linear search complexity, which can become costly as dataset size and query volume increase [4]. Modern vector databases therefore employ approximate nearest neighbour (ANN) indexes such as inverted file index (IVF) and hierarchical navigable small world index (HNSW), which can reduce retrieval latency, with index configuration affecting the trade-off between search speed and retrieval quality [4, 18]. More recent approaches, including DiskANN and SPANN, further improve scalability by storing large portions of the index on SSDs while maintaining lightweight routing structures in memory, enabling efficient retrieval over billion-scale datasets [19, 20]. These approaches substantially improve vector search performance but focus on optimising retrieval within individual vector stores rather than across heterogeneous and distributed knowledge stores.

2.5 Relational Search

While semantic retrieval addresses unstructured information, enterprise knowledge

also resides within relational databases. Relational systems typically employ B-tree indexes for range queries, hash indexes for equality comparisons, while PostgreSQL also provides GIN indexes for full-text and JSON search [4, 21]. Kleppmann and Riccomini [4] note that although indexes improve retrieval performance, they increase storage requirements and slow write operations because they must be maintained as the underlying data changes. Consequently, index selection represents a trade-off between query performance and maintenance overhead.

2.6 Distributed Systems

As enterprise knowledge continues to grow, a single machine may no longer be sufficient to store or process all data. Distributed database systems therefore introduce concerns such as sharding and replication. The degree of fragmentation directly influences query performance [22], while the choice of sharding strategy depends heavily on workload characteristics. Key-range partitioning is effective when queries exhibit locality, whereas hash-based partitioning generally provides a more balanced distribution of reads and writes [4]. Similar principles apply to vector databases, where semantically similar vectors may be clustered together to reduce the search space during retrieval [23]. Distributed systems also introduce opportunities for parallel query execution; however, Özsu and Valduriez [24] note that reducing response time through parallelism may increase overall execution time because of additional communication overhead. Consequently, computation should occur as close to the data as possible whenever feasible [24].

2.7 Query Planning

These considerations ultimately influence query planning within distributed retrieval systems. Özsu and Valduriez [25] argue that query optimisation should minimise communication costs by reducing the amount of data transferred between nodes. They recommend assigning execution costs to alternative query plans and favour strategies that perform selection as early as possible, thereby reducing intermediate results and avoiding expensive join operations wherever possible [25]. Such observations reinforce the importance of orchestration strategies that intelligently coordinate retrieval across multiple heterogeneous knowledge stores.

3. Methodology

This project followed an iterative software-development methodology. The aim was to design, implement, and evaluate a working distributed heterogeneous retrieval system rather than to conduct a purely observational research study.

Development was divided into small, testable increments so that each layer could be verified before the next was added. The main increments were data acquisition, schema design, shard construction, relational retrieval, vector retrieval, orchestration, end-to-end integration, and benchmarking.

An iterative approach was selected because several design decisions depended on evidence obtained during implementation. For example, the final data distribution could not be confirmed until the movie corpus had been collected, and valid selective routes could not be fixed until the shard layouts were known. Implementing one complete path at a time also reduced the risk of building a large system whose components worked independently but failed when integrated.

The methodology combined four related activities:

4. Results and Analysis

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5. Evaluation

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6. Conclusion

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References

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Appendices

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